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BIO 101 — Past Questions & Materials

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Paper 1 of 2 — BIO 101 slides- Growth and Development

Level 100 • First Semester • Handout

BIO 101

General Biology I

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1

Growth and Development

■ Growth is defined as the irreversible increase in size and dry weight of an organism. The word irreversible is used because growth is a permanent change.

■ Growth is a characteristic of all living things. Non-living things do not grow.

Growth is brought about by three processes,

✓ Cell division: The cells divide by mitosis to increase in number.

✓ Cell elongation: The new cells produced in cell division elongate causing an increase in the size of the organism.

✓ Cell differentiation: The cells then undergo structural and physiological changes in order to be suited for a particular function in the organism. During differentiation, the cells get specialized to perform particular functions in the organism.

Growth and Development

■ Development: This is increase in complexity of an organism. After its birth, the baby grows to increase in size. It also develops and becomes more complex in structure and functioning.

■ Increase in height and weight in a baby is growth but starting to walk, talk, and presence of teeth is development. Growth and development occur co-currently within the organism.

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Measurement of growth.

■ The rate of growth can be estimated by measuring a particular growth parameter over a period of time.

■ The parameter chosen should be suitable for that particular organism. In bacteria, growth is estimated by counting the number of cells in a given time.

■ In man and other mammals, growth is estimated by measuring linear dimension and weight. Two types of weight are recognized when estimating growth

✓ Fresh weight/mass. This is the mass of a living organism including all its

moisture content. It is an easy way to estimate growth and it does not involve killing of the organism. However, it gives crude results and is not reliable.

✓ Dry weight/mass. This is the mass of an organism after all moisture is removed. This gives an accurate measure of growth but it involves killing of the organism during dehydration. It is also not easy to carry out.

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Growth curve

■ This is a graph which shows the change of a given growth parameter with time. This graph is S- shaped in most living organisms and it is called the sigmoid curve

■ The curve shows 4 phases.

✓ Lag phase: This is a period of slow growth. It is the first phase of growth where there are very few cells dividing and the organism is getting used to the environment.

✓ The exponential phase: This is a phase of rapid growth. It is the second phase where the cells dividing are many and the organism is used to the environment.

✓ Decelerating growth phase. This is a period where growth slows down. The deceleration in growth may be due to; 1. Competition for food, space and other resources. 2. The organism is preparing for reproduction. 3. The organism is aging

Growth curve

Growth curve

✓ The plateau phase. /stationary phase: This is a period where there is no change in the growth parameter under investigation. At this point the number of cells, which die is equal to those produced. After the plateau, the growth decelerates in seasonal organism due to aging and dispersal. In perennial organisms' growth increases continuously.

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Factors affecting growth in animals and plants

✓ Nutrients. The availability of nutrients affects growth. All organisms require nutrients to build up new cells and tissues. They also require energy to grow. The plants make their own food in photosynthesis but also absorb nutrients such as nitrates, phosphates, e.t.c from the soil to manufacture certain components. Absence of these may hinder growth. Animals feed on already made food, which they digest to obtain the nutrients necessary for growth.

✓ Light: Green plants require light to carry out photosynthesis in order to grow. Animals, fungi and bacteria may not require light to grow but light affects their activities in different ways.

✓ pH: Some organisms grow under a narrow range of pH. When the pH changes, growth of the organisms is affected. pH changes in the soil affect the absorption of minerals in the soil. pH also affects the action of enzymes. 9

Factors affecting growth in animals and plants

✓ Temperature: There is usually an optimum temperature for an organism in which growth occurs favorably. Temperature affects the action of enzymes in the body; it affects germination of seeds and uptake of water in plants. All these have a bearing in growth of plants and animals.

✓ Accumulation of toxic wastes: Through their metabolism, organisms produce toxic wastes. When these accumulate, they affect the growth of organisms. Most organisms excrete these wastes and some convert them to less toxic substances

✓ Hormones: Hormones are also known to control growth in organisms. Such hormones are called growth hormones/ growth substances. In humans, the human growth hormone and thyroxine control growth while in plants growth is controlled by auxins.

✓ Genes Given all other conditions favorable, the final size to which an organism grows depends on genes that are inherited from its parent. The genes also determine all the characteristics of that organism

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Growth in flowering plants

■ In plants, growth does not take place all over the body of the plant but in particular regions called meristems.

■ Meristems are regions whose cells have the ability to divide and produce new cells.

■ There are three types of meristems in plants.

✓ Apical meristems. These are found at the growing tips that is shoot tip and root tips. They divide to produce new cells and the plant increases in linear dimensions (height/length). This is called primary growth. 2.

✓ Lateral meristems. These are found in the region of permanent tissue. They include the vascular cambium and cork cambium. Their division results into increase in girth (circumference) of the plant and it is called secondary growth

Growth in flowering plants

✓ Intercalary meristems: These are located at the base of the shoot of some monocot plants. It results into primary growth in other areas other than the tips. Intercalary meristems allow regeneration of the plants susceptible to damage e.g. grasses.

■ There are two types of growth in plants.

✓ Primary growth: This involves germination and the activity of the apical meristems to produce shoots and roots. It results into increase in linear

dimension.

■ Secondary growth. This occurs in woody dicot plants. It results from the activity of lateral meristems and causes a plant to increase in girth. Secondary growth is also called secondary thickening. During secondary growth, the vascular cambium divides to produce new phloem and xylem. Secondary growth provides more support to the plant, as it grows taller

Seed germination

■ Germination is the development of an embryo into a seedling under favorable conditions. There are two types of germination.

✓ Epigeal germination. This is a type of germination where the hypocotyl elongates faster than the epicotyls pushing the cotyledons above the ground. (hypocotyl is portion of radicle below the cotyledons). This type of germination occurs in dicotyledonous plants.

✓ Hypogeal germination This is a type of germination where the epicotyl elongates faster than the hypocotyl pushing the cotyledons below the ground. (Epicotyls are the region of radicle above the cotyledons).

✓ This type of germination occurs in monocotyledonous plants. During epigeal germination the seed absorbs water through the micropyle in a process called imbibition.

Conditions necessary for germination

■ Seeds are dispersed in a dormant state and they require favorable conditions to germinate. These conditions are;

✓ Water: Water is needed to hydrolyse food reserves, activate enzymes, to act as a medium in which all enzymes act and a medium for transport of dissolved food substances.

✓ Optimum temperature There is an optimum temperature, which is favorable for a particular seed. Temperature affects the activity of enzymes involved in germination. Below 0°C germination does not occur because the enzymes are inactive and at very high temperatures the enzymes are denatured and protoplasm is destroyed.

✓ Air: Air contains oxygen, which is necessary for respiration to provide energy. Seeds do not germinate in cooled and boiled water because it contains no air.

Seed dormancy.

■ This is a period of no growth in a seed where the metabolic activity of the seed is greatly reduced. In this state, seeds cannot germinate even under favorable conditions.

■ Causes of dormancy.

✓ Lack of adequate water and warmth.

✓ Presence or absence of light, soil pH and germination inhibitors.

✓ Hard impervious testa, which is impermeable to water.

- ✓ Immaturity of embryo.
- ✓ Presence of germination inhibitors e.g., abscisic acid

Methods of breaking dormancy

- ✓ Providing suitable conditions for germination.
 - ✓ Passing seeds through guts of animals to weaken the seed coat by enzymes.
 - ✓ Alternating heat and cooling of seeds may result into cracking the seed coat.
 - ✓ Allowing the embryo time to mature.
 - ✓ Using growth promoters such as gibberellins.
- It allows dispersal of seeds in time and space.
 - It ensures that germination occurs in favorable conditions to allow the seedlings to grow.
 - It allows time for the embryo in a seed to fully mature.
 - It allows food storage in seeds and their preservation as food.

Paper 2 of 2 — BIO 101 General Biology I

Level 100 • First Semester • Handout

BIO 101

General Biology I

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1

Respiration

- Respiration is a metabolic process that occurs in all organisms. It is a biochemical process that occurs within the cells of organisms.
- In this process, nutrients are broken down into useful energy (ATP-Adenosine triphosphate) which is further used by cells to perform various functions.
- Every living species, from a single-celled organism to dominant multicellular organisms, performs respiration.
- Respiration can also be defined as the biochemical activity of the cell in which glucose is broken down in a series of reaction to release chemical energy known as ATP, Carbondioxide (C) and Water for Aerobic or ATP, Alcohol and Water for Anaerobic.
- There are two phases of respiration
 - ✓ Gaseous Exchange(External respiration)
 - ✓ Cellular respiration (Internal respiration or tissue respiration)

External Respiration

- GASEOUS EXCHANGE (EXTERNAL RESPIRATION): All organism that respire aerobically (i.e with oxygen), takes in oxygen into the respiratory organ (inhalation) and gives out carbon dioxide (exhalation) as waste product.
- The oxygen taken in oxidizes the food, and this leads to the formation of carbondioxide. The exchange of these two gasses is known as external respiration (breathing). And the process by which these gasses are exchanged is by diffusion.
- Gaseous exchange could be direct or indirect.
- Direct External Respiration: this is a type of respiration in which dissolved oxygen from the surrounding(water or wet) diffuses directly into the cell/cells and carbondioxide diffuses out directly without any special respiratory system required.

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External Respiration

- Simple organism like spirogyra, Amoeba, Hydra e.t.c undergoes direct

respiration. Indirect External Respiration: Because organism evolve to be more complex, direct respiration can no longer work for the high oxygen demand. And hence specialized structures found within the animal's body are developed for Indirect Respiration to take place. And these may include: Lungs (reptiles, birds, Mammals), Gills (fishes and crabs), Tracheal tubes (Insects), Moist Skin (earthworms, toads and frogs) e.t.c In indirect respiration not all cell are in contact with the environment

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TYPES OF RESPIRATORY SURFACES, ORGANISM AND THE ENVIRONMENT IN WHICH THEY FUNCTION.

S/N Respiratory surface Organisms Functioning environment

1 Cell membrane Amoeba and Aquatic

Paramecium

(microscopic aquatic organism)

2 Gills Fishes, Tadpoles, Aquatic

Crustacea. etc.

3 Skin Frogs and Toads, Aquatic

Earthworm etc..

4 Trachea Insects Terrestrial

5 Lungs Reptiles, Birds, Terrestrial

Mammals

6 Mouth Frogs and Toads Terrestrial

7 Stomata Leaves Terrestrial

8 Lenticels Stems Terrestrial

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Tissue Respiration

■ This is the breakdown of food substances to release energy. It occurs with the help of enzymes. The major food respired (respiratory substrate) is a carbohydrate (glucose).

■ All other compounds are converted into a carbohydrate before they are respired. The energy released is stored as ATP (Adenosine tri phosphate).

■ ATP is highly energy rich compound formed between a chemical bond between ADP (Adenosine di phosphate) and inorganic phosphate groups, i.e. $ADP + P_i \rightarrow ATP$.

■ If the energy stored as ATP is required by the body, ATP is suddenly broken down into ADP and P_i to release energy for the body activities i.e. $ATP \xrightarrow{ATPase} ADP + P_i + \text{energy}$

Tissue Respiration

- The energy released is used by the body for various activities i.e
 - ✓ Maintaining blood circulation
 - ✓ Bring about breathing movement
 - ✓ For producing sound Transmission of nerve impulses from one part to another.
 - ✓ Synthesis of blood proteins
 - ✓ Maintaining the constant blood temperature
 - ✓ Cell division either mitosis or meiosis leading to growth
 - ✓ Active transport of materials into or outside the cell.
 - ✓ Secretion of various materials like hormones, enzymes, etc.

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Chemical Processes of Cellular Respiration

- Cellular respiration requires series of chemical reactions and respiratory enzymes.
 - ✓ Glycolysis: Partial oxidation of a glucose molecule to form 2 molecules of pyruvate. This process takes place in the cytosol.
 - ✓ Formation of Acetyl CoA: Pyruvate formed in glycolysis enters the mitochondrial matrix. It undergoes oxidative decarboxylation to form two molecules of Acetyl CoA. The reaction is catalysed by the pyruvate dehydrogenase enzyme.
 - ✓ Krebs cycle (TCA cycle or Citric Acid Cycle): It is the common pathway for complete oxidation of carbohydrates, proteins and lipids as they are metabolised to acetyl coenzyme A or other intermediates of the cycle. The Acetyl CoA produced enters the Tricarboxylic acid cycle or Citric acid cycle. Glucose is fully oxidized in this process.

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like NADH and FADH₂.

Chemical Processes of Cellular Respiration

The acetyl CoA combines with 4-carbon compound oxaloacetate to form 6C citrate. In this process, 2 molecules of CO₂ are released and oxaloacetate is recycled. Energy is stored in ATP and other high energy compounds.

- ✓ Electron Transport System and Oxidative Phosphorylation: ATP is generated when electrons are transferred from the energy-rich molecules like NADH and FADH₂, produced in glycolysis, citric acid cycle and fatty acid oxidation to molecular O₂ by a series of electron carriers. O₂ is reduced to H₂O. It takes place in the inner membrane of mitochondria.

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Glycolysis

- This is also known as sugar- breaking process. It is the metabolic process that breaks down or convert glucose into pyruvic acid.
- Glycolysis is the process in which glucose is broken down to produce

energy. It produces two molecules of pyruvate, ATP, NADH and water. The process takes place in the cytoplasm of a cell and does not require oxygen.

- It occurs in both aerobic and anaerobic organisms. Glycolysis is the primary step of cellular respiration, which occurs in all organisms. Glycolysis is followed by the Krebs cycle during aerobic respiration. In the absence of oxygen, the cells make small amounts of ATP as glycolysis is followed by fermentation.

Glycolysis Pathway

- The glycolysis pathway occurs in the following stages:
- Stage 1: A phosphate group is added to glucose in the cell cytoplasm, by the action of enzyme hexokinase. In this, a phosphate group is transferred from ATP to glucose forming glucose,6- phosphate.
- Stage 2: Glucose-6-phosphate is isomerised into fructose,6-phosphate by the enzyme phosphoglucumutase.
- Stage 3: The other ATP molecule transfers a phosphate group to fructose 6-phosphate and converts it into fructose 1,6-bisphosphate by the action of the enzyme phosphofructokinase.
- Stage 4: The enzyme aldolase converts fructose 1,6-bisphosphate into glyceraldehyde 3- phosphate and dihydroxyacetone phosphate, which are isomers of each other.

Glycolysis Pathway

- Step 5: Triose-phosphate isomerase converts dihydroxyacetone phosphate into glyceraldehyde 3-phosphate which is the substrate in the successive step of glycolysis.
- Step 6: This step undergoes two reactions: The enzyme glyceraldehyde 3-phosphate dehydrogenase transfers 1 hydrogen molecule from glyceraldehyde phosphate to nicotinamide adenine dinucleotide to form $\text{NADH} + \text{H}^+$. Glyceraldehyde 3-phosphate dehydrogenase adds a phosphate to the oxidised glyceraldehyde phosphate to form 1,3-bisphosphoglycerate.
- Step 7: Phosphate is transferred from 1,3-bisphosphoglycerate to ADP to form ATP with the help of phosphoglycerokinase. Thus two molecules of phosphoglycerate and ATP are obtained at the end of this reaction.
- Step 8: The phosphate of both the phosphoglycerate molecules is relocated from the third to the second carbon to yield two molecules of 2-phosphoglycerate by the enzyme phosphoglyceromutase

Glycolysis Pathway

- Step 9: The enzyme enolase removes a water molecule from 2-phosphoglycerate to form phosphoenolpyruvate.
- Step 10: A phosphate from phosphoenolpyruvate is transferred to ADP to form pyruvate and ATP by the action of pyruvate kinase. Two

molecules of pyruvate and ATP are obtained as the end products.

Krebs's cycle

- It is an eight-step process. Krebs cycle or TCA cycle takes place in the matrix of mitochondria under aerobic condition.
- Step 1: The first step is the condensation of acetyl CoA with 4-carbon compound oxaloacetate to form 6C citrate, coenzyme A is released. The reaction is catalysed by citrate synthase.
- Step 2: Citrate is converted to its isomer, isocitrate. The enzyme aconitase catalyses this reaction.
- Step 3: Isocitrate undergoes dehydrogenation and decarboxylation to form 5C α -ketoglutarate. A molecular form of CO₂ is released. Isocitrate dehydrogenase catalyses the reaction. It is an NAD⁺ dependent enzyme. NAD⁺ is converted to NADH.
- Step 4: α -ketoglutarate undergoes oxidative decarboxylation to form succinyl CoA, a 4C compound. The reaction is catalyzed by the α -ketoglutarate dehydrogenase enzyme complex. One molecule of CO₂ is released and NAD⁺ is converted to NADH.

Krebs's cycle

- Step 5: Succinyl CoA forms succinate. The enzyme succinyl CoA synthetase catalyses the reaction. This is coupled with substrate-level phosphorylation of GDP to get GTP. GTP transfers its phosphate to ADP forming ATP.
- Step 6: Succinate is oxidised by the enzyme succinate dehydrogenase to fumarate. In the process, FAD is converted to FADH₂.
- Step 7: Fumarate gets converted to malate by the addition of one H₂O. The enzyme catalysing this reaction is fumarase.
- Step 8: Malate is dehydrogenated to form oxaloacetate, which combines with another molecule of acetyl CoA and starts the new cycle. Hydrogens removed, get transferred to NAD⁺ forming NADH. Malate dehydrogenase catalyses the reaction.

DIFFERENCES BETWEEN GLYCOLYSIS AND KREBS' CYCLE

Glycolysis Krebs's cycle

It is the first step of respiration yielding two molecules Krebs Cycle is the second step of aerobic respiration in

of pyruvic acid after the partial breakdown of a which pyruvate is oxidized completely into inorganic glucose molecule in a set of enzymatic processes substances forming carbon dioxide

Occurs in all the living organisms Occurs in aerobes

Occurs inside the cytoplasm Occurs inside the mitochondria

No carbon dioxide evolved Carbon dioxide evolved

Oxygen not required for glycolysis Oxygen is required for Krebs Cycle

Four ATP molecules are produced in the glycolysis for One ATP or GTP molecule is produced by substrate-

each glucose molecule level phosphorylation in each turn of the Krebs cycle

Consumes 2 molecules of ATP for initial Doesn't consume ATP

phosphorylation of substance molecules

Net gain of two molecules of ATP and two molecules Each turn of the Krebs cycle yields three molecules of

of NADH gained for every molecule of glucose broken NADH and two molecules of FADH₂

down